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DETAIL SPECIFICATION

INDICATOR SOLUTION, pH DETERMINATION (RANGE 3.0 TO 11.0)

Inactive for new design after 28 May 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an indicator solution for the determination of pH values of water in the range of 3.0 to 11.0.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-537 - Disks, Color Standards

Comments, suggestions, or questions on this document should be addressed to DLA Aviation, VEB, 6090 Strathmore Rd, Richmond, VA 23297 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)

APHA SME - Standard Methods for the Exam of Water and Wastewater

(Copies of these documents are available at https://www.standardmethods.org.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

Research Paper 2619 - pH Values of the Clark and Lubs Buffer Solutions at 25ºC, by Vincent E. Bower and Roger G. Bates

(Research Paper 2619 was published in the Journal of Research of the National Bureau of Standards, Volume 55, No. 4, October 1955, pgs. 197-200. Copies of the journal may be obtained online at https://www.nist.gov/nist-research-library or by e-mail from Library@nist.gov.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The indicator solution shall produce a color for specific pH values of water that visually match color standards of a master color disk. The solution shall be suitable for use in determining pH values of water from 3.0 to 11.0 in increments of 1 pH when tested in accordance with 4.2.

3.2 Standard sample. A standard sample of indicator solution will be supplied by the Government on request by a potential bidder. The standard sample will be furnished only to show the potential bidder the color (obtained when 0.5 milliliter (mL) of standard sample is added to 15 mL of each buffer solution prepared in accordance with 4.2.2) that the indicator solution he proposes to furnish must produce (see 6.4).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Suspended matter and foreign material. Total suspended matter and foreign material shall not exceed 10 mg/L when tested per 4.2.1. When specified (see 6.2), the contractor may certify that the indicator solution has been filtered through Whatman grade 934AH and 984H, Gelmen type A/E, Millipore type AP40, or equivalent glass fiber filters prior to packaging.

3.4 Government-loaned property. Unless otherwise specified (see 6.2), the following property conforming to MIL-D-537, type I, class 8, in the quantities indicated will be loaned by the Government (see 6.3):

Item No.	Description	Quantity
1	Disk, master color standard, medium values, pH 3.0 to 11.0	1
2	Disk, master color standard, low values, pH 3.0 to 11.0	1
3	Disk, master color standard, high values, pH 3.0 to 11.0	1
4	Disk evaluator	1

4. VERIFICATION

4.1 Sampling. Sampling shall be accordance with MIL-STD-1916, verification level II or as specified in the acquisition document (see 6.2).

4.2 Tests.

4.2.1 Suspended matter foreign material. Unless otherwise specified, samples selected in accordance with 4.1 shall be tested for suspended matter and foreign material in the solution in accordance with APHA “Standard Methods for the Examination of Water and Wastewater”, section 2540D “Total Suspended Solids dried at 103 - 105ºC”. See 3.3.1 for limits of suspended matter and foreign material.

4.2.2 Color. Buffer solutions of pH 3.0 to 11.0 inclusive in increments of 1 pH shall be prepared in accordance with NBS Research Paper 2619. 0.50 mL of indicator shall be added to 15 mL of each buffer solution. The mixtures shall be compared for color with the Government-loaned low, medium and high master color standard disks. The color of each pH mixture shall be visually in the range of color of the low and high master color standard disks as per 3.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the

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managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicator solution covered by this specification is intended for use in determining the pH of water by comparison with color disks conforming to MIL-D-537, type I, class 8.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When Government-loaned property is not required (see 3.4)
c. Alternate sampling requirements (see 4.1)
d. When certification is acceptable (see 3.3.1)
e. Unit size required (i.e. 4 or 16 ounces)
f. Packaging requirements (see 5.1)
g. Special marking requirements
h. Activities requiring a copy of the SDS (see 6.5)

6.3 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.4, when applicable (see 6.2).

6.4 Standard sample. The contracting officer should arrange to furnish potential bidders a 16 ounce bottle of indicator solution, pH wide range, NSN 6810-00-087-2340.

6.5 Safety Data Sheets (SDS). Contracting officers will identify those activities requiring copies of completed Safety Data Sheets prepared in accordance with FED-STD-313 and meeting the requirements of 29 CFR 1910.1200. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313. 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the SDS (see 6.2).

6.6 Subject term (key word) listing.

Chemical solution

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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